Universal Anomaly Detection for Nonparametric Data

Intern report to be submitted
by

Medavarapu Atchutha Rama Murthy
21EC39016

Under the guidance of

Dr. Jithin Ravi



ELECTRONICS AND ELECTRICAL COMMUNICATION ENGINEERING

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Department of Electronics and Electrical
Communication Engineering
Indian Institute of Technology,
Kharagpur
India - 721302

CERTIFICATE

This is to certify that **Medavarapu Atchutha Rama Murthy** (**21EC39016**), a student of the **Department of Electronics and Electrical Communication Engineering**, has carried out a project under my supervision during the period **10th May 2025 to 15th July 2025**. The project was entitled **Universal Anomaly Detection for Nonparametric Data**. The student has shown good analytical ability, sincerity, and commitment in completing the assigned work. I am confident that the skills demonstrated will be valuable in his future endeavors. The undersigned may be contacted for any further information regarding this work.

Supervisor

Department of Electronics and Electrical Communication Engineering
Indian Institute of Technology,
Kharagpur

Place: Kharagpur
Date: September 8, 2025

ABSTRACT

We study the problem of anomaly detection within the framework of universal outlier hypothesis testing. The goal is to identify sequences that do not conform to the typical distribution without prior knowledge of the outlier model. Our work focuses on nonparametric tests that rely on distributional distances rather than parametric assumptions. In particular, we analyze kernel-based approaches using the Maximum Mean Discrepancy (MMD) and information-theoretic measures like the Kullback–Leibler (KL) divergence) for distinguishing anomalous data. Theoretical guarantees are provided in terms of error exponents, consistency, and asymptotic optimality.

To validate the theoretical results, we conduct simulations on synthetic data, with Gaussian distributions as typical benchmarks, examining detection accuracy, error probabilities, and stopping criteria. Our study highlights the trade-offs between threshold selection, sample complexity, and computational efficiency in sequential and fixed-sample settings. The results show that kernel methods and divergence measures provide reliable tools for anomaly detection in both structured and unstructured data.

Keywords: Anomaly Detection, Hypothesis Testing, Universal Outlier Problem, Maximum Mean Discrepancy, KL Divergence, Error Exponent, Nonparametric Methods, Sequential Testing, Gaussian Distribution, Stopping Time, Threshold

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Chapter 1

Introduction

1.1 Problem Overview

We study the problem of identifying outlier sequences from a group of data streams. Each stream gives i.i.d. samples over time. All but few streams follow the same unknown distribution, while one or few streams (outliers) follows a different unknown distribution. The challenge is to detect those as early as possible, with high accuracy, and without knowing any distribution in advance.

We focus on the *universal, nonparametric, and sequential* setting:

- **Universal**: No assumptions about the actual distributions.
- **Nonparametric**: Detection relies solely on observed data
- **Sequential**: Decisions are made as data arrives over time.

1.2 Motivation

In real-world systems, distributions of data are often unknown or can change over time. Fixed-sample (batch) methods are not suitable for applications requiring fast or online decisions. A test that is both sequential and nonparametric is more practical and flexible. Also, existing methods may perform well only under strong assumptions. Our goal is to study tests that work universally and still give good performance in terms of speed and accuracy. This motivates the use of kernel-based methods like MMD^2 and error exponent analysis.

1.3 Applications and Broader Impact

This problem appears in many applications:

- **Cybersecurity**: Detecting an intruder among multiple network flows.
- **Sensor networks**: Identifying faulty or noisy sensors.
- **Healthcare**: Finding abnormal patient signals from wearable devices.
- **Quality control**: Spotting defective machines in production lines.

The broader impact includes building efficient and real-time systems for monitoring and detection without needing full knowledge of the system model.

1.4 Current Experiments and Evaluation Metrics

We simulate the sequential anomaly detection test (NP-SEQ) using synthetic data. Each stream is generated under either the nominal or outlier distribution. Samples are observed over time, and the test decides when to stop and which stream is the outlier.

The performance is evaluated using:

- **Misidentification probability** $\mathbb{P}_{\mathcal{I}}[\delta \neq \mathcal{I}]$
- **False alarm probability** $\mathbb{P}_0[\delta \neq \emptyset]$
- **Stopping time** τ
- **Error exponent**: rate of decay of error with time or number of samples

We also study how test parameters (like α, C, kernel choice) affect performance.

Chapter 2

Literature Review

2.1 Overview of Outlier Detection

Outlier detection aims to identify data sequences that deviate from the majority population. In the classical setting, all but a few streams are generated from an unknown nominal distribution p, while anomalous streams follow a different unknown distribution q, with $p \neq q$. The challenge lies in developing distribution-free tests that reliably distinguish outliers without prior knowledge of p or q. Applications include sensor fault detection, cyber-security, and health monitoring.

2.2 Universal Hypothesis Testing Framework

Universal hypothesis testing focuses on statistical tests that are effective without prior knowledge of the underlying data distributions. The goal is to identify anomalous data streams from a group of otherwise homogeneous streams. A key metric for a test's performance is the **maximal error probability**, which measures the worst-case probability of misidentifying an anomaly.

2.2.1 Generalized Likelihood (GL)

The Generalized Likelihood (GL) test is a method for outlier detection in a universal, nonparametric setting where the distributions of normal and anomalous data are unknown. This approach identifies the most likely anomalous stream by maximizing a generalized likelihood function. This is equivalent to a more intuitive test based on measuring the distance between the empirical distributions of the data streams.

The GL test can be expressed as minimizing a sum of distances between the empirical distributions. Specifically, when neither the nominal nor the anomalous distribution is known, the test identifies the outlier by:

$$\delta(y^{Mn}) = \underset{i=1,\ldots,M}{\operatorname{argmin}} \sum_{j \neq i} D\left(\gamma_j \parallel \frac{\sum_{k \neq i} \gamma_k}{M-1}\right). \quad (1)$$

This formulation effectively identifies the stream that is most dissimilar to the collective group of all other streams.

2.2.2 Error Exponents in Hypothesis Testing

The **error exponent** characterizes the exponential rate at which the probability of error decreases as the sample size increases. Formally, for a maximal error probability $P_{\max}(n)$ with n samples, the error exponent is defined as:

$$E = -\lim_{n \to \infty} \frac{1}{n} \log P_{\max}(n). \quad (2)$$

A higher error exponent means the test is more powerful and reliable, as the error probability decays faster. A test is considered **universally exponentially consistent** if its error exponent remains positive regardless of the underlying data distributions, ensuring its effectiveness in a wide range of scenarios without prior knowledge.

2.2.3 Observations on Optimal Error Exponents

From our analysis of the literature, we've observed that the optimal error exponent for the maximal error probability is characterized by the Bhattacharyya distance.

Bhattacharyya distance ($B(\mu, \pi)$) is a measure of the similarity between two discrete or continuous probability distributions. It is closely related to the Bhattacharyya coefficient, and for probability distributions μ and π, it is defined as:

$$B(\mu, \pi) = -\ln\left(\sum_x \sqrt{\mu(x)\pi(x)}\right).$$

The distance ranges from 0 to infinity, where 0 indicates identical distributions and larger values indicate greater divergence. In this context, it quantifies how different the nominal distribution is from the anomalous one.

Based on the research, we found that when both the nominal and anomalous distributions are known, the optimal exponent is equal to $2B(\mu, \pi)$. This result

is also achievable by the GL test when only the anomalous distribution is known. Furthermore, it is interesting to note that even when only the mean of the nominal distribution is known, one can still achieve this optimal exponent using a different test.

However, a key observation from the papers is that it is not possible to design a single test that universally achieves this optimal error exponent when neither distribution is known. This suggests that a completely universal test must accept a lower performance ceiling, which is an important consideration for designing robust, real-world anomaly detection systems.

2.3 Kernel Methods; Maximum Mean Discrepancy

Kernel methods provide a way to measure distributional differences without explicit density estimation. The Maximum Mean Discrepancy (MMD) between two distributions p and q is defined as:

$$\mathrm{MMD}^2(p, q) = \mathbb{E}_{x,x' \sim p}[k(x, x')] + \mathbb{E}_{y,y' \sim q}[k(y, y')] - 2\,\mathbb{E}_{x \sim p, y \sim q}[k(x, y)], \quad (10)$$

where $k(\cdot, \cdot)$ is a positive definite kernel (e.g., Gaussian kernel).

Given finite samples $X_i^n = \{x_{i1}, \ldots, x_{in}\}$ and X_j^n, the unbiased empirical estimator is:

$$M_u^2(i, j, n) = \frac{1}{n(n-1)} \sum_{l \neq m} h(z_l, z_m), \quad (11)$$

where $z_l = (x_{il}, x_{jl})$ and $h(z_l, z_m) = k(x_{il}, x_{im}) + k(x_{jl}, x_{jm}) - k(x_{il}, x_{jm}) - k(x_{im}, x_{jl})$.

2.4 Sequential Nonparametric Outlier Detection

This section introduces the problem of detecting anomalous data streams in a sequential setting, as described in a relevant research paper [1]. This approach builds upon the fixed-sample-size and generalized likelihood frameworks previously discussed.

2.4.1 Problem Formulation

The authors of [1] address the challenge of detecting anomalous data streams from a finite set of S streams. The key assumption is that a small number, L, of these streams are anomalies, originating from an unknown distribution q, while the remaining $S - L$

streams follow an unknown nominal distribution p. The distributions p and q are distinct, but no further information about them is available. The paper focuses on two specific cases: when there is **exactly one anomaly** ($L = 1$), and when there are **at most** A **anomalies** ($0 \leq L \leq A < S/2$).

2.4.2 The Proposed NP-SEQ Test

A **universal, distribution-free sequential test** called **NP-SEQ** is proposed in [1]. This test is designed to make a decision at each time step as new samples arrive, leveraging the **Maximum Mean Discrepancy (MMD)** as its core decision statistic.

The test statistic, a **max-min MMD distance** (Γ_n), is a crucial component. At each time step n, a candidate anomalous subset $\hat{I}_n$ is identified by finding the group of streams that maximizes the minimum MMD distance between themselves and the rest of the streams:

$$\hat{I}_n = \underset{A \in S_A}{\operatorname{argmax}} \min_{i \in A} \min_{j \in S \setminus A} M_u^2(i, j, n). \quad (3)$$

Here, $M_u^2(i, j, n)$ is the unbiased empirical estimator of the squared MMD between stream i and stream j. The test statistic is then defined as the max-min distance for the chosen subset:

$$\Gamma_n = \min_{i \in \hat{I}_n} \min_{j \in S \setminus \hat{I}_n} M_u^2(i, j, n). \quad (4)$$

The sequential test stops and declares a decision when this statistic exceeds a time-varying threshold, $T_n = C/n^\alpha$. The parameters C and α are critical. C is a constant that controls the test's sensitivity and the tradeoff between stopping time and error probability; a larger C generally leads to a lower error but a longer stopping time. The exponent α dictates how quickly the threshold decays over time.

2.4.3 Performance Guarantees and Consistency

The paper provides significant theoretical guarantees for the proposed test. The NP-SEQ test is proven to be **universally consistent**, meaning the probability of error goes to zero as the confidence parameter C increases, ensuring the test is reliable for any distinct pair of distributions p and q. This property is a key advantage of the test's nonparametric approach.

Chapter 3

Related Works

3.1 Implementation of Sequential Test [1]

This section outlines the logical structure of the implemented simulations for the sequential nonparametric anomaly detection tests. The core algorithms are designed to evaluate the performance of different distribution distance metrics and test scenarios.

3.1.1 MMD Computation

The implemented MMD computation calculates the unbiased empirical squared Maximum Mean Discrepancy between two finite data streams, which serve as empirical distributions. The logic relies on a Radial Basis Function (RBF) kernel to measure pairwise distances within and between the two data sets, with the final MMD value serving as a measure of their dissimilarity.

```
Function MMD_COMPUTATION(Data_X, Data_Y, Sigma):
  n = number of samples in Data_X
  m = number of samples in Data_Y

  // Calculate kernel matrices
  Kernel_XX = exp(-pairwise_distance(Data_X, Data_X)^2 / (2 * Sigma^2))
  Kernel_YY = exp(-pairwise_distance(Data_Y, Data_Y)^2 / (2 * Sigma^2))
  Kernel_XY = exp(-pairwise_distance(Data_X, Data_Y)^2 / (2 * Sigma^2))
```

```
  // Exclude diagonal elements for unbiased estimation
  Set diagonal of Kernel_XX to 0
  Set diagonal of Kernel_YY to 0

  // Compute MMD^2 value
  MMD_squared = (sum(Kernel_XX) / (n * (n-1))) +
                (sum(Kernel_YY) / (m * (m-1))) -
                (2 * sum(Kernel_XY) / (n * m))

  Return MMD_squared
```

3.1.2 KL Divergence Computation

In addition to MMD, a test based on Kullback-Leibler (KL) Divergence was implemented for comparative analysis. The logic computes the KL divergence between two data streams by first estimating their probability distributions from histograms. This provides an alternative nonparametric metric to assess the difference between the streams.

```
Function KL_DIVERGENCE_COMPUTATION(Data_X, Data_Y, Bins):
  // Estimate probability distributions from histograms
  px = histogram(Data_X, Bins, density=True)
  py = histogram(Data_Y, Bins, density=True)

  // Add small constant to avoid log(0)
  px = px + 1e-10
  py = py + 1e-10

  // Compute KL divergence: sum(px * log(px / py))
  KL_value = sum(px * log(px / py))

  Return KL_value
```

3.1.3 Sequential Anomaly Detection Test (NP-SEQ)

The central logic of the NP-SEQ test operates sequentially. At each time step, one new sample is collected from each of the S data streams. The test then computes a distance metric (either MMD or KL Divergence) for each stream against the combined set of all other streams. The stream with the largest distance is identified as the candidate anomaly. The test continues until this largest distance value exceeds a decaying threshold, at which point a final decision is made and the stopping time is recorded.

```
Function NP_SEQ_TEST(Alpha, C, Metric_Type, Anomaly_Mean, ...):
  For each simulation trial:
    Initialize empty streams and set anomaly index
    For n from 2 to Max_Steps:
      Collect one sample from each stream
      For each stream i:
        Combine all other streams into 'Others'
        If Metric_Type is MMD:
          Metric_Value[i] = MMD_COMPUTATION(Stream[i], Others, ...)
        Else if Metric_Type is KL:
          Metric_Value[i] = KL_DIVERGENCE_COMPUTATION(Stream[i], Others, ...)

      Candidate_Anomaly = stream with max Metric_Value
      Threshold = C / (n ^ Alpha)

      If Metric_Value[Candidate_Anomaly] > Threshold:
        Log stopping time n
        Log error (if Candidate_Anomaly is incorrect)
        Break inner loop

    If loop completes without stopping:
      Log max_steps as stopping time and record an error

  Calculate average stopping time and error probability
```

3.1.4 Simulating Different Scenarios

To evaluate the test's robustness, the core NP-SEQ logic was applied to various simulation settings.

3.1.4.1 Varying Mean

The NP-SEQ test is run with different mean values for the anomalous distribution. This logic tests how the test's efficiency and accuracy change as the separation between the anomalous and typical distributions increases. The simulation iterates through a predefined list of mean values and records performance metrics for each.

3.1.4.2 Misidentification for Multiple Anomalies

The test was extended to a scenario with a known number of multiple anomalies. The logic was adapted to identify a candidate set of anomalies (the top A streams with the largest distance values) and measure the probability of misidentification—that is, whether the identified set of anomalies correctly matches the true set.

3.1.4.3 Non-Gaussian Distributions

To demonstrate the test's universality, a simulation was performed using a uniform distribution for both typical and anomalous streams. This confirms that the nonparametric approach is not limited to Gaussian data. The sample generation function is modified to produce uniform data, while the core test logic remains the same.

3.1.4.4 Testing the Decay Rate (α)

In addition to the scenarios above, we conducted simulations with a range of α values beyond those noted in the original paper. This was done to empirically observe the impact of the threshold's decay rate on test performance. The results provide insight into how the choice of α influences the trade-off between the test's efficiency (expected stopping time) and its accuracy (error probability).

3.2 Simulation Results

3.2.1 Gaussian Distribution Implementation Results

The following figures present the results of the NP-SEQ test for detecting an anomalous stream from a Gaussian distribution. The visualizations compare the test's performance across various parameter settings.

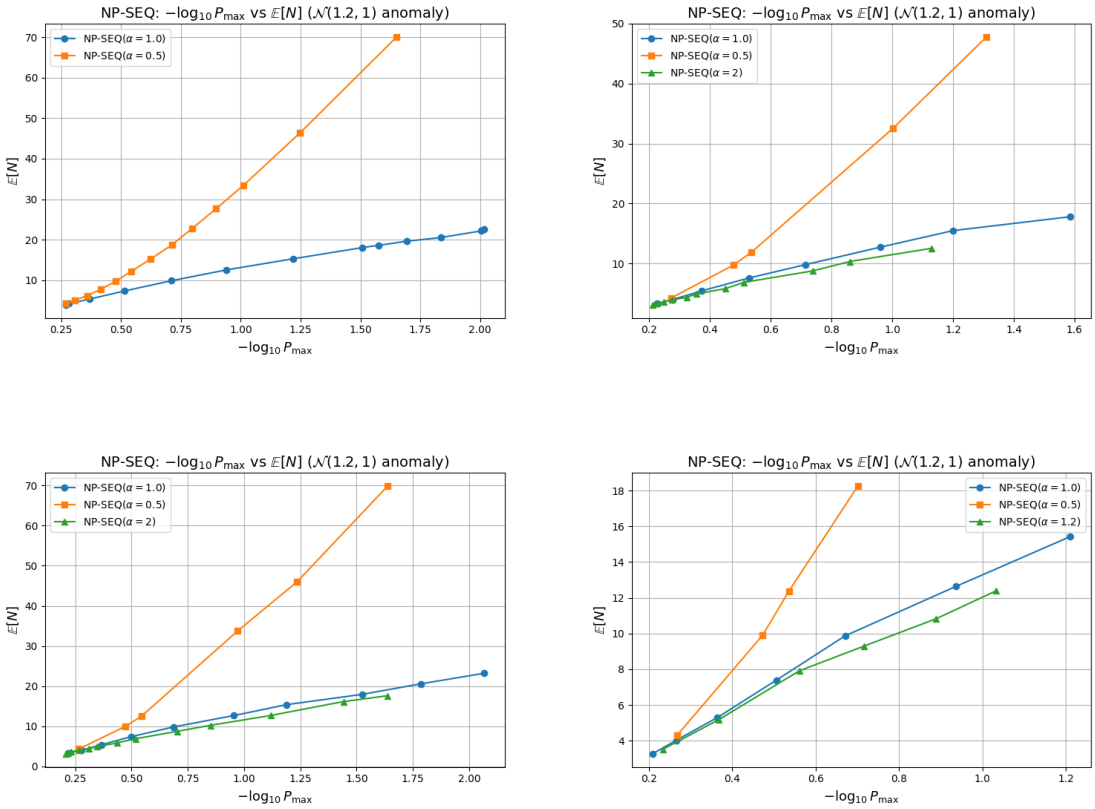


Figure 3.1: MMD comparison plots for Gaussian distributions with varying parameters.

- The plots show a clear trade-off between the expected stopping time ($\mathbb{E}[N]$) and the maximal error probability ($P_{\max}$). To achieve a lower error probability, the test requires more samples, leading to a higher $\mathbb{E}[N]$.
- As the confidence parameter, C, increases, the threshold for the test also increases, resulting in a higher observed expected stopping time, $\mathbb{E}[N]$.

3.2.2 Gaussian Distribution with Multiple Means

This section presents the results of the NP-SEQ test when the anomalous stream's mean is varied. The typical streams are from a standard normal distribution, while the anomalous stream's mean is shifted to evaluate how the degree of separation between the distributions affects the test's performance.

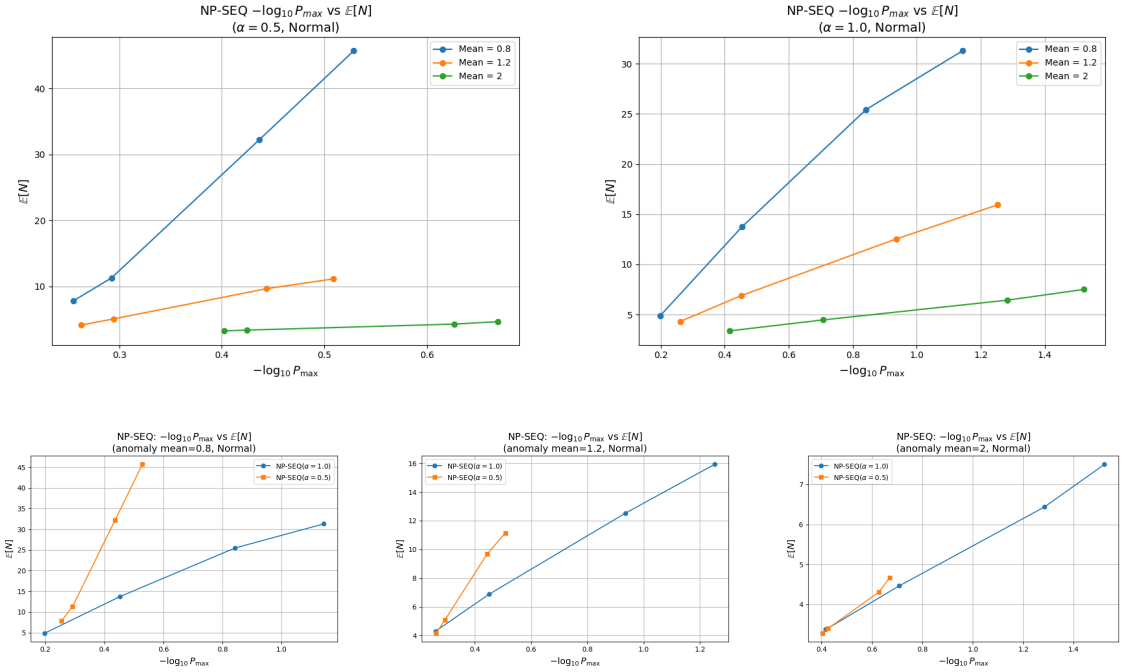


Figure 3.2: MMD performance plots for different anomalous distribution means.

- As the mean of the anomalous distribution increases, the expected stopping time ($\mathbb{E}[N]$) of the test decreases. This is because a larger mean difference leads to a greater separation between the distributions, making the anomaly easier to detect with fewer samples.

- This observation confirms the intuition that MMD-based tests are more efficient at detecting more pronounced anomalies. The test requires fewer samples to cross the detection threshold when the difference between the typical and anomalous distributions is more significant.

3.2.3 KL Divergence Implementation Results

This section presents the results of the sequential test using Kullback-Leibler (KL) Divergence as the primary metric. The simulations evaluate the test's performance with a single anomaly and, in a separate case, with varying anomalous distribution means, for comparison against the MMD-based approach.

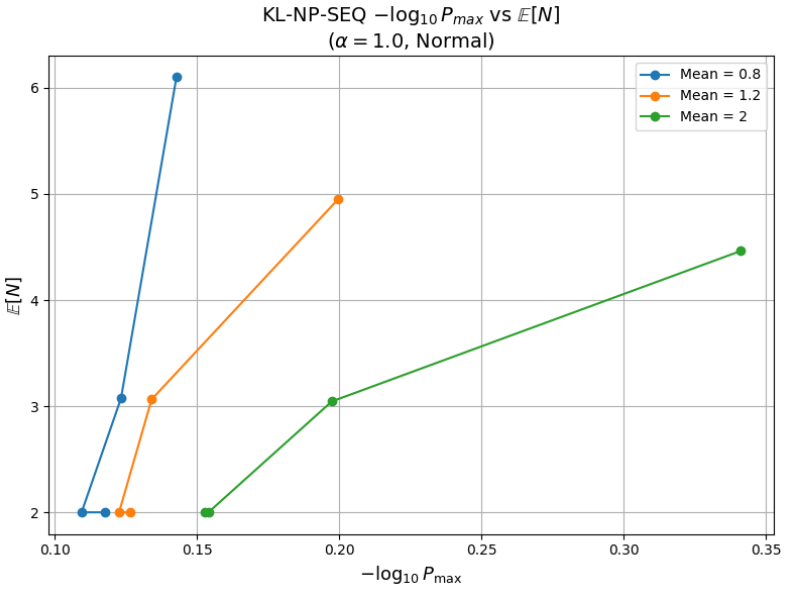


Figure 3.3: KL Divergence performance plot showing the relationship between expected stopping time and error probability for various parameter settings.

- The simulation results using KL Divergence show a behavior similar to that observed with the MMD metric, confirming the accuracy of the sequential test logic with a different nonparametric metric.
- As the mean of the anomalous distribution increases, the expected stopping time ($\mathbb{E}[N]$) of the test decreases. This confirms that the test is more efficient at detecting anomalies that are more pronounced or "farther away" from the typical distribution.

3.2.4 Misidentification for Multiple Anomalies

This section presents the simulation results for the misidentification test, where the test's ability to correctly identify a known number of multiple anomalies is evaluated. The test was run with a fixed $\alpha = 0.5$ and an anomalous mean of 1.2. The following figures compare the performance for the one-anomaly and two-anomaly cases.

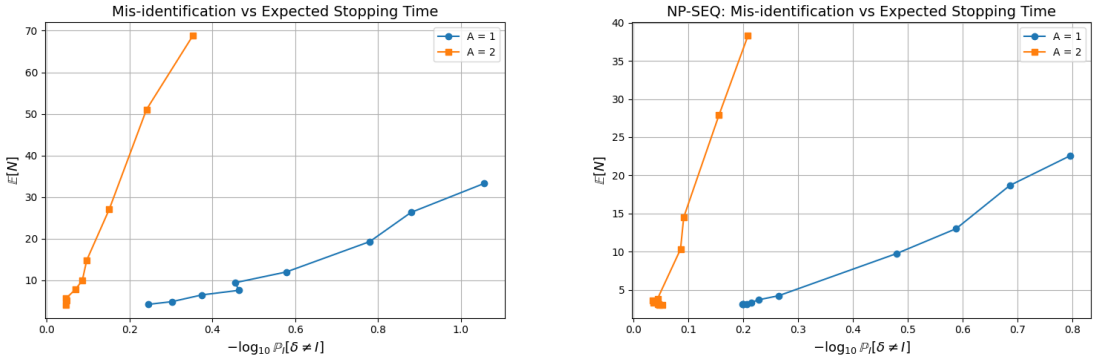


Figure 3.4: Misidentification vs. Expected Stopping Time for the one- and two-anomaly cases.

- The results show that for a given expected stopping time ($\mathbb{E}[N]$), the probability of misidentification for the two-anomaly case is higher compared to the one-anomaly case.
- This indicates that detecting two anomalies simultaneously is more challenging than detecting a single anomaly. The increased complexity of identifying the correct set of outliers requires a more conservative threshold or more samples to achieve a low error rate, leading to a higher $\mathbb{E}[N]$ for the two-anomaly case.

3.2.5 Uniform Distribution Implementation Results

To validate the "universal" and nonparametric nature of the NP-SEQ test, further simulations were conducted using non-Gaussian data. For this section, the typical streams were drawn from a standard uniform distribution, $p = \mathcal{U}(0, 1)$, while the anomalous stream was drawn from a shifted uniform distribution, $q = \mathcal{U}(\mu, \mu + 1)$. This setup tests the algorithm's performance on distributions without a central tendency and evaluates how the degree of separation (controlled by μ) impacts detection efficiency.

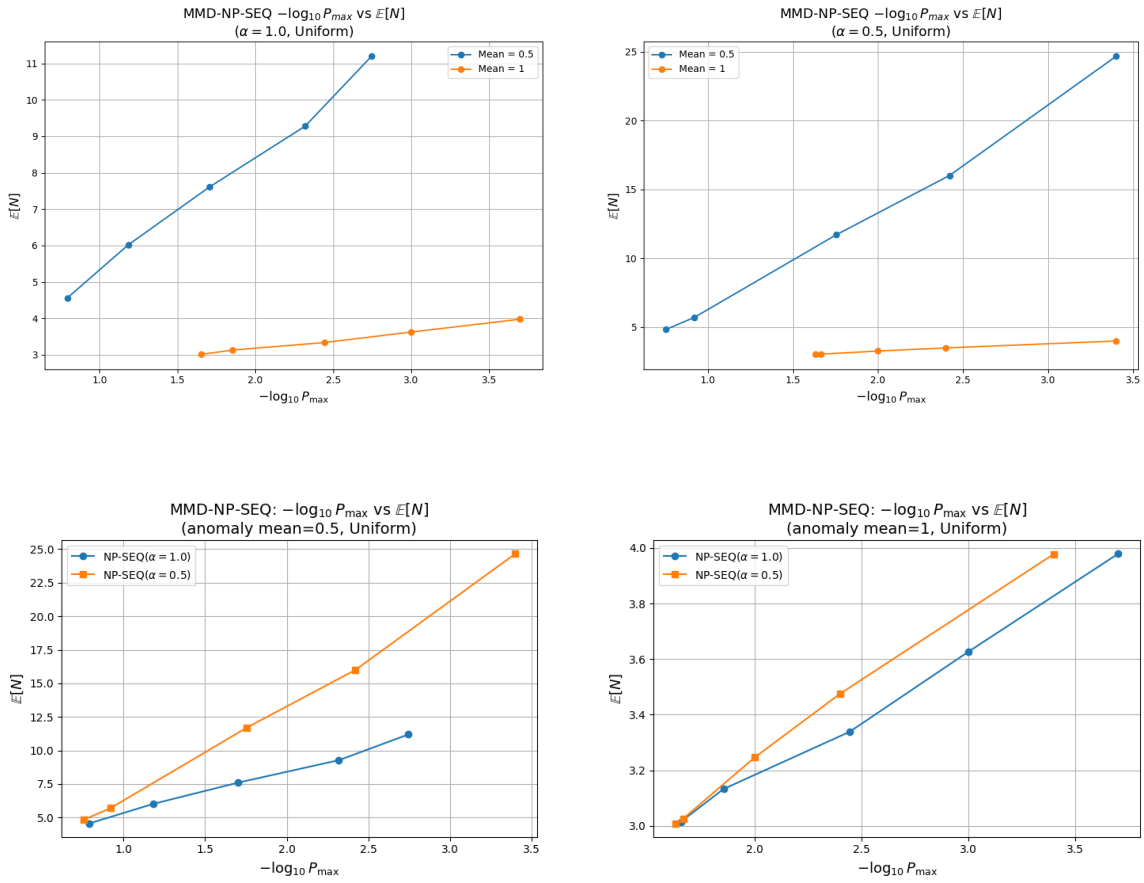


Figure 3.5: MMD-NP-SEQ performance on uniform distributions for varying parameters.

- The plots confirm the test's effectiveness on non-Gaussian data, demonstrating its nonparametric utility. The observed trends are consistent with the results from the Gaussian simulations.
- Increasing the anomalous mean from $\mu = 0.5$ to $\mu = 1.0$ is observed to significantly reduce the expected stopping time, $\mathbb{E}[N]$. More distinct anomalies are detected faster.
- A threshold decay rate of $\alpha = 1.0$ consistently results in a lower expected stopping time for a given error probability when compared to $\alpha = 0.5$.
- Across all simulations, a clear trade-off is visible between the stopping time and the error rate: achieving a lower error probability requires a higher $\mathbb{E}[N]$.

3.3 Observations and Conclusions

This section summarizes the main findings from the different simulation tests and presents the final conclusions drawn from the results.

3.3.1 Key Observations

From the various experiments, several key patterns about the test's behavior were consistently observed:

- **The Test is Truly Nonparametric:** A central observation is that the test works well on completely different types of data. It showed similar, predictable performance for both bell-shaped (Gaussian) and flat (Uniform) distributions. This confirms that the method is effective without needing any prior knowledge of the data's shape.

- **Detection Speed Depends on Anomaly Difference:** The single biggest factor affecting performance was how different the anomalous data was from the normal data. When the anomaly's mean was shifted further away, making it more obvious, the test stopped much faster ($\mathbb{E}[N]$ dropped significantly). This was true for all data types and metrics used.

- **A Faster Threshold Decay is More Efficient:** In all head-to-head comparisons, the threshold decay rate of $\alpha = 1.0$ was more efficient than $\alpha = 0.5$. It consistently led to a lower expected stopping time for the same level of accuracy, making it a better parameter choice in these scenarios.

- **The Sequential Logic is Robust:** The core step-by-step logic of the test is solid and independent of the specific metric used. When the MMD metric was replaced with KL-Divergence, the overall behavior of the test remained the same—faster detection for more obvious anomalies and the same trade-offs between speed and accuracy.

- **More Anomalies Make the Problem Harder:** The test for multiple anomalies showed that finding two anomalies is clearly more difficult than finding one. To achieve the same low error rate, the test needed more samples, which resulted in a higher expected stopping time.

3.3.2 Conclusions

The simulation results lead to several important conclusions about the practical use and effectiveness of this anomaly detection method:

- **The Method is Validated:** The implemented sequential test performs as theoretically promised in the reference paper [1]. The results successfully show its ability to find anomalies in a universal setting and confirm the fundamental trade-off between detection speed and the probability of error.
- **Sequential Testing is Highly Efficient:** A major conclusion is the sample efficiency of the sequential approach. For anomalies that are easy to spot, the test can stop very early, saving significant time and computational resources compared to a method that must always analyze a large, fixed-size batch of data. This makes it very practical for real-time monitoring.
- **Performance is Tunable and Predictable:** The test is not a "black box." Its performance is directly tied to clear parameters. The results show that the test's speed is a function of the anomaly's distinctiveness and the choice of the threshold decay rate, α. This provides a clear understanding of how to tune the test for a given application's needs.

3.4 Sequential Detection Using Rank Stability

The detection of multiple anomalies presents a significant challenge, as calculating a reliable test statistic for a group of streams is complex. A potential simplification is a **rank-based sequential test**, which shifts the focus from the exact numerical anomaly scores to their relative ordering over time. The core intuition is that anomalous streams should **consistently rank higher** than typical streams. While the scores of normal streams might fluctuate randomly, the high ranks of the anomalous streams should show a persistent pattern.

A possible implementation of this idea would be to rank all streams at each time step based on an anomaly score like MMD. The test would then stop not when a score crosses a threshold, but when the set of the top-ranked streams remains **stable for a certain number of consecutive steps**. This approach would be simpler and potentially more robust to random noise, as it relies on identifying a consistent pattern rather than a single high value.

Chapter 4

Future Works

This study provides a foundation for several promising research directions to deepen the understanding and expand the capabilities of sequential anomaly detection.

- **Theoretical Analysis of the Decay Rate (α):** The simulations consistently showed that a decay rate of $\alpha = 1.0$ was more efficient than $\alpha = 0.5$ or $\alpha = 2.0$. A formal theoretical investigation could explain this empirical result, analyzing the optimal trade-off between a threshold that decays quickly enough to be efficient but slowly enough to avoid errors from early-stage noise.

- **Threshold Optimization and False Alarm Control:** The threshold parameters, C and α, were set manually. A significant area for future work is to develop methods for optimizing these parameters. This could involve creating a model to select the best threshold based on a user's desired false alarm rate, or even designing an adaptive threshold that adjusts itself to the data in real-time.

- **Application to Discrete Distributions:** The experiments in this project focused on continuous data. A valuable next step would be to apply the MMD-based test to discrete data (data from a finite alphabet). This would allow for a direct performance comparison against classical methods like the Generalized Likelihood (GL) test, which was specifically designed for that type of data.

- **Performance on High-Dimensional Data:** The current tests were on one-dimensional data. A crucial extension would be to evaluate the method on high-dimensional datasets. This is where the MMD metric is expected to shine, and it would test how well the approach scales to more complex, real-world problems.

- **Development of the Rank-Based Test:** The proposed rank-based test offers a simpler, more intuitive way to handle multiple anomalies. A key future project is to fully implement and evaluate this method, comparing its detection power and computational cost against the more complex group-based statistics.

- **Empirical Study of the GL Test Exponent:** The reference paper [2] shows the GL test can achieve an optimal error exponent related to the Bhattacharyya distance. An interesting study would be to implement the fixed-sample GL test and empirically find the conditions under which it achieves this optimal performance, providing a valuable benchmark for the sequential method.

- **Benchmarking Against Modern Methods:** To position this work in the broader field of machine learning, the NP-SEQ test could be benchmarked against other popular anomaly detection algorithms, like Isolation Forests or deep learning-based autoencoders, on a standard set of test data.

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